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PrismData

B01 - NLP Credit Score Development with Prism Data Utilizing Cashflow Underwriting for Fairer Credit Assessment

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Introduction

Traditionally, credit scores rely heavily on financial history, focusing on assets, liabilities, and net worth. This method can disadvantage low-income individuals, new graduates, contract workers, and immigrants who may lack extensive credit records despite being financially responsible. Our method addresses this gap by analyzing income and transaction data to develop a more comprehensive understanding of financial behavior. Cash flow underwriting expands credit access and enables a more inclusive and accurate assessment of creditworthiness.

Datasets

Key datasets used to assess consumer creditworthiness:

- **Account Data** – Consumer account types, balances, and balance dates.
- **Consumer Data** – Consumers' delinquency (binary, where 1 = delinquent).
- **Transaction Data** – Transaction categories, amount, credit/debit indicators
- **Categories** – Maps transaction category IDs to category names

Data Exploration

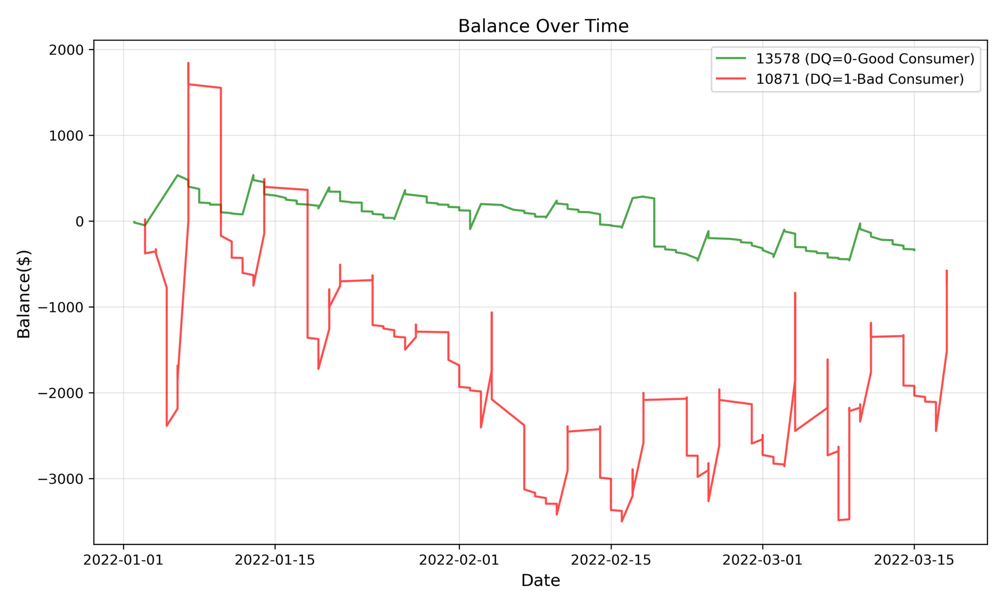

Figure 1: Delinquent and non-delinquent consumer trend-Non-DQ maintain higher running balance overall.

Non-delinquent borrowers tend to have a **higher median and mean paycheck amount** compared to delinquent borrowers.

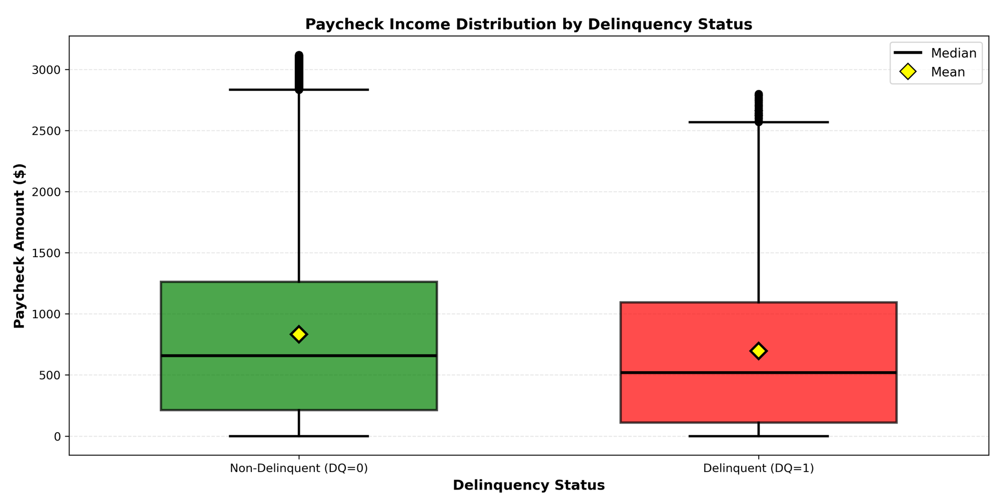

Figure 2: Paycheck variability based on delinquency status.

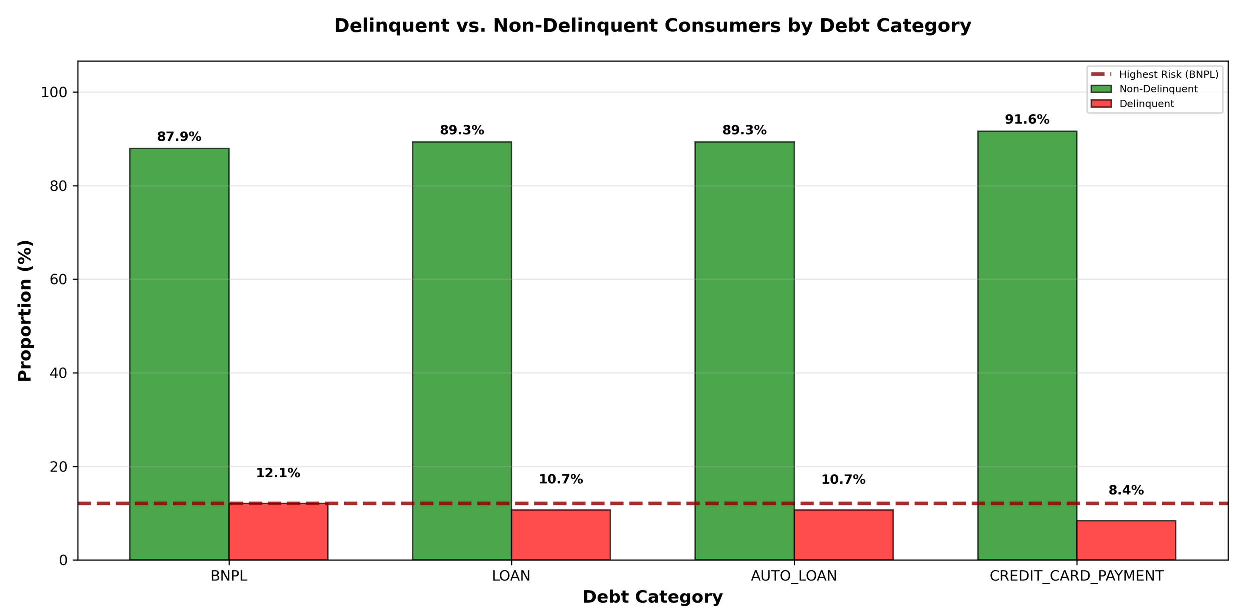

Figure 3: Delinquency rate among debt payment categories.

Model Pipeline

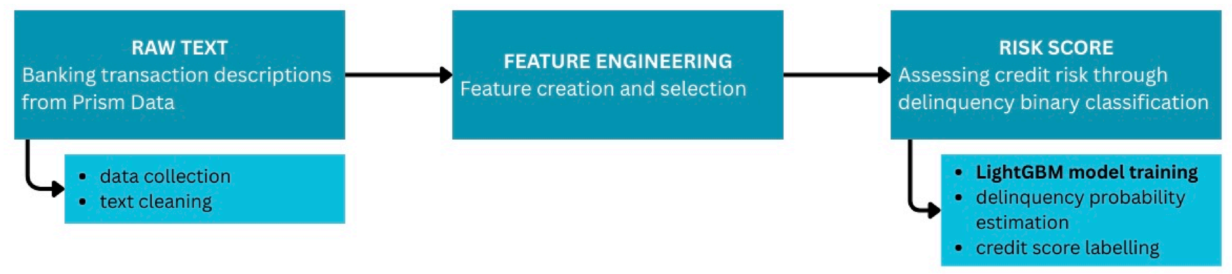

Figure 4: Model pipeline: from raw data ingestion through feature engineering, model training (Logistic Regression, Random Forest, XGBoost, LightGBM), and evaluation.

Model Evaluation - Metrics used to assess model performance:

- **ROC-AUC:** Ability to distinguish between delinquent and non-delinquent borrowers; higher values indicate better performance.
- **Accuracy:** Percentage of borrowers correctly classified overall.
- **Precision:** Proportion of predicted delinquent borrowers who actually become delinquent.
- **Recall:** Proportion of actual delinquent borrowers correctly identified.

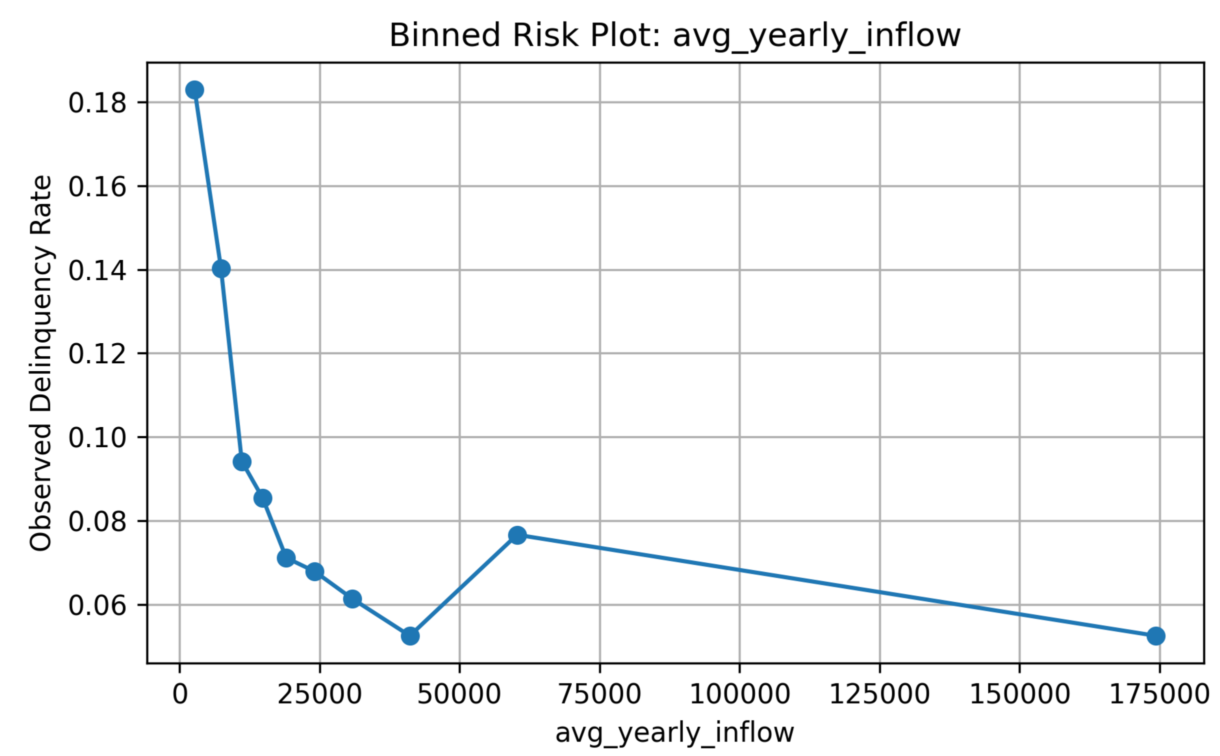

Figure 5: Negative trends in delinquency rates when binning consumers into 10 bins by average yearly inflow.

Feature Engineering

- **Transaction Aggregation:** monthly balance summaries (mean, min, max), credit/debit ratios, net flow, income stability (yearly inflow, standard deviation, trend slope), income span
- **Category-Based:** per-category spend totals and averages across 48 categories (groceries, rent, ATM cash, etc.) + proportion features like `pct_spend_essentials` and `paycheck_ratio`
- **Time Windows:** Time-windowed behavioral flags — 6-month indicators for overdraft, BNPL usage, and account fees; spending volatility; and a discretionary drop flag (30%+ decline in entertainment/travel/fitness spend)

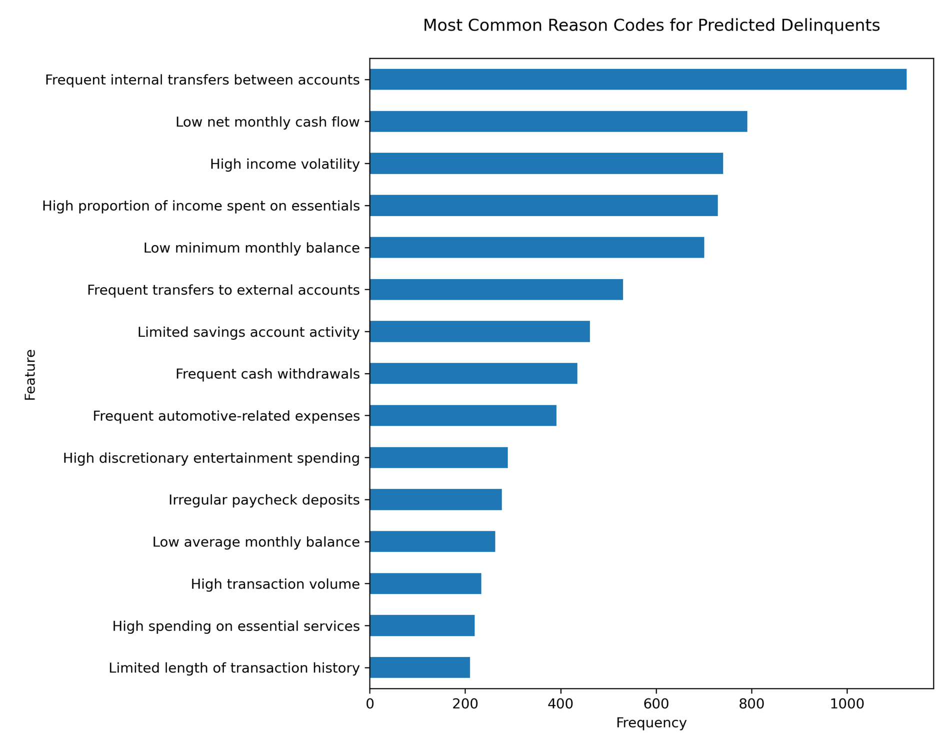

Figure 6: Negative trends in delinquency rates when binning consumers into 10 bins by average yearly inflow.

50 features selected via XGBoost feature importance optimizing ROC-AUC to **0.801**

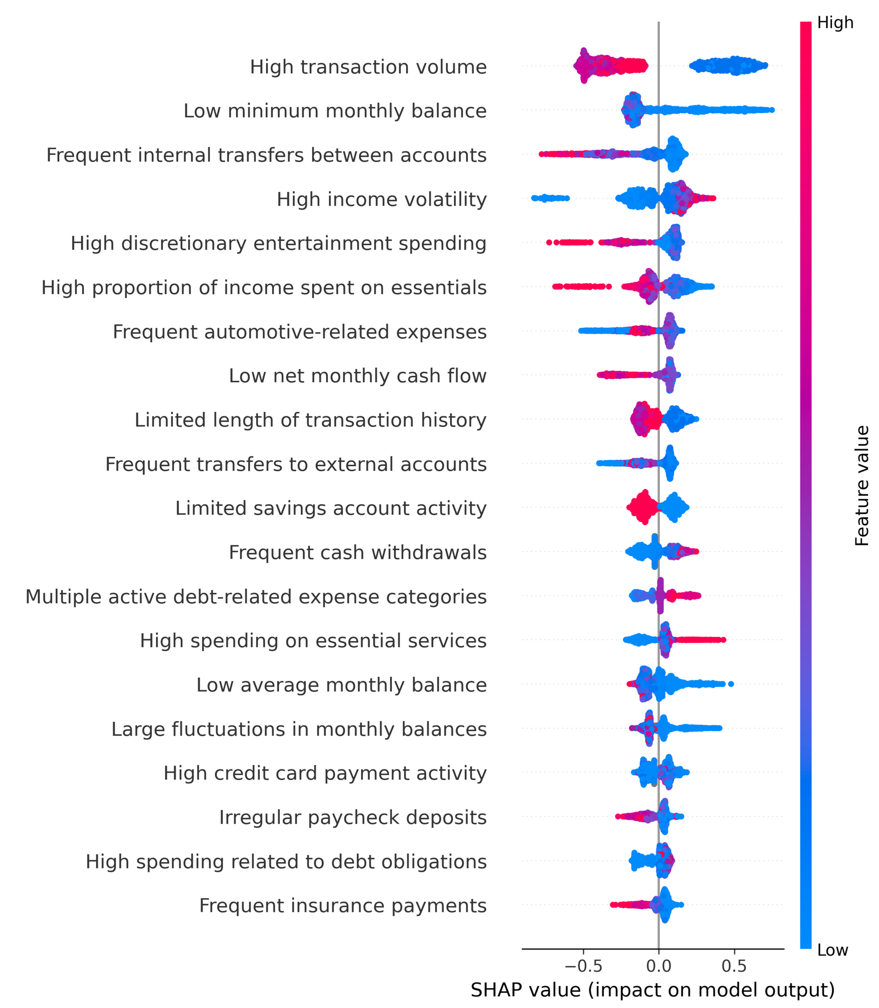

Figure 7: SHAP plot showing the top 20 most impactful features in the delinquency prediction model.

Findings

- **Credit card payments** and **loan activity** were among the strongest predictors, suggesting debt obligation behavior is highly indicative of delinquency risk
- Higher average monthly inflow was associated with lower delinquency risk
- **Overdraft** history is a strong risk signal: consumers with an overdraft in the past 6 months were more than twice as likely to become delinquent compared to those without
- Spending patterns across categories add predictive power in predicting delinquency by significantly raising AUC during forward selection
- Consumers with **near-zero net monthly flow** are higher risk than those with negative months: people with very negative months likely have higher incomes and financial flexibility, while those barely breaking even have little buffer against financial shocks

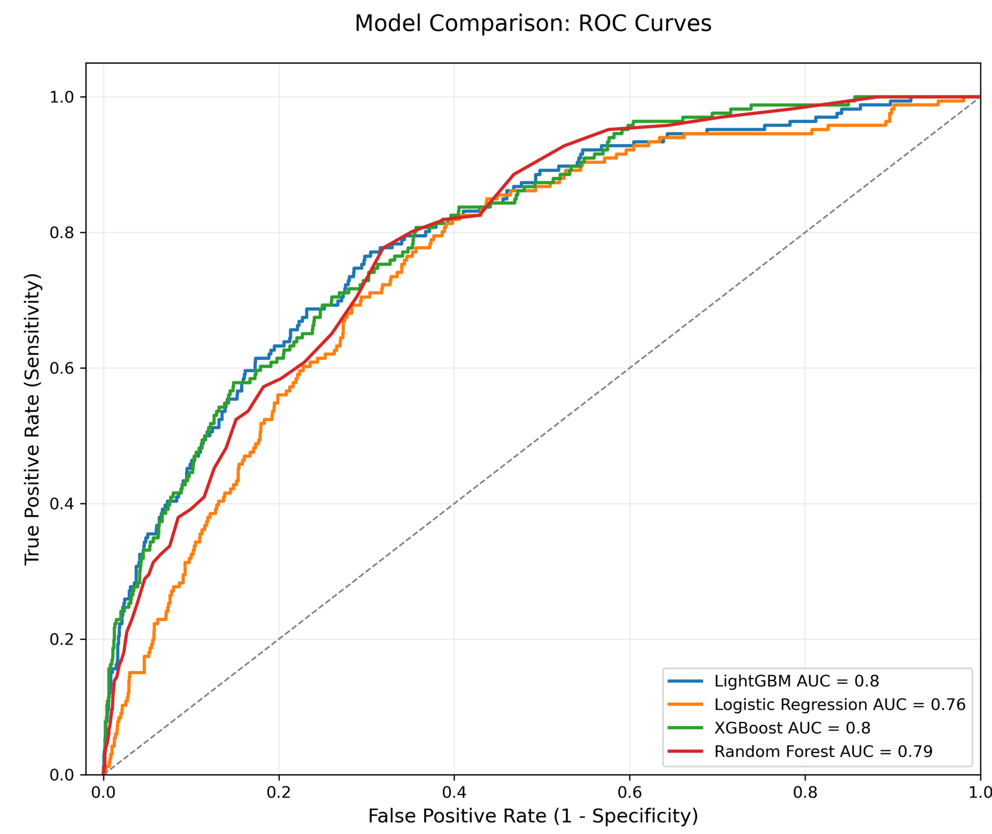

Figure 8: ROC Curve for Logistic Regression, Random Forest, XGBoost, and LightGBM

Next Steps

- **Model Improvements:** Address class imbalance in delinquency through resampling or cost-sensitive learning
- **Feature Development:** Explore longer time windows for behavioral flags beyond the existing 6-month indicators
- **Fairness & Bias:** Audit model predictions across demographic groups to ensure the cashflow approach genuinely reduces bias compared to traditional credit scoring